THE EVALUATION MAP OF A MODULE INTO ITS DOUBLE DUAL IS INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Pontryagin dual of a module with respect to Q/Z and the evaluation map into the double dual. defined immediately before the statement, source lines 13794-13802. Supplies the map whose injectivity is asserted.
- The abelian group Q/Z is injective. stated in the proof at proof lines 13809-13810 and used at lines 13814-13816. Extends the homomorphism defined on a cyclic subgroup to all of M, which is what converts the local nonvanishing into a statement about the evaluation map.
- Existence of a nonzero homomorphism from a nonzero cyclic group to Q/Z. used at proof lines 13811-13813. Provides the functional that does not annihilate the chosen element x.

Source reference. Original lemma 01DB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1792–1792. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *For any R-module M the evaluation map $ev : M \to (M^\vee)^\vee$ is injective.*

Proof. You can check this using that $\mathbf{Q}/\mathbf{Z}$ is an injective abelian group. Namely, if $x \in M$ is not zero, then let $M' \subset M$ be the cyclic group it generates. There exists

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a nonzero map $M' \to \mathbf{Q}/\mathbf{Z}$ which necessarily does not annihilate x. This extends to a map $\varphi : M \to \mathbf{Q}/\mathbf{Z}$ and then $ev(x)(\varphi) = \varphi(x) \neq 0$. $\square$